Electrocardiogram Signal–Based Cardiac Abnormality Detection Using Machine Learning

1 Introduction

Electrocardiogram (ECG) is one of the most commonly used and important physiological signals in clinical practice, which is widely used for the early screening and monitoring of heart disease [1]. Automated analysis of ECG signals has great significance in the field of biomedical engineering and electrical engineering, as manual interpretation is not only time-consuming but also influenced by subjective factors. Due to the prevalence of noise, baseline drift, and individual variations in ECG signals, anomaly detection based on real electrocardiogram signals remains a challenging engineering problem. This project aims to investigate methods based on the machine learning for detecting cardiac abnormalities in ECG signals, and to compare the performance of different modeling strategies under realistic physiological noise conditions.

2 Problem Specification

This project models the detection of abnormal electrocardiograms as a supervised learning problem. Given an input ECG signal segment $x(t)$, our goal is to determine whether the corresponding cardiac activity is normal or abnormal in the presence of noise, individual variability and signal non-stationarity. This problem is defined as a classification task. We focus on comparing two different machine learning modeling: classical machine learning models relying on manual features versus neural network models capable of automatically learning representations from ECG signals. The core is: different modeling approaches differ in terms of performance and generalization capability under real-world, noisy physiological signal conditions.

3 Data

This project uses the MIT-BIH Arrhythmia Database, a widely used dataset for ECG analysis [2]. The dataset consists of 48 half-hour ECG recordings collected from 47 subjects, sampled at 360 Hz using two ECG channels. Over 110,000 heartbeats are annotated by cardiologists, providing reliable labels. The dataset includes real noise and physiological variability, making it suitable for evaluating real-world performance. ECG signals are segmented into heartbeat windows and preprocessed through normalization and filtering prior to model training.

4 Approach

One model is based on a classical machine learning approach, where ECG features will be extracted from each heartbeat segment to capture time-domain and morphological characteristics. Principal Component Analysis (PCA) will be applied for dimensionality reduction, along with a Support Vector Machine (SVM) classifier for heartbeat classification. This model emphasizes interpretability and feature-based learning. Another model adopts a neural network approach, in which a one-dimensional Convolutional Neural Network (1D-CNN) will be trained directly on raw ECG signal segments [3]. The model will incorporate batch normalization and dropout layers to improve generalization and reduce overfitting. This approach enables automatic feature learning from raw data without manual feature engineering [4].

5 Evaluation Criteria

Model performance will be evaluated using classification metrics including accuracy, precision, recall, and the F1 score [5]. A confusion matrix will be used to analyze true and false classifications, with particular emphasis on minimizing false negatives due to their importance in medical applications. These metrics will be used to compare the classical and neural network approaches under realistic ECG signal conditions. Since the model is applied in a medical context, it should aim to achieve an accuracy of at least 95%.

6 Conclusions

We explored machine learning–based detection of cardiac abnormalities from ECG signals. Comparing classical and neural network models highlights performance and robustness trade-offs. The project emphasizes practical considerations such as noise, data variability, and regularization.

References

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PROBLEM

- Electrocardiogram (ECG) signals are widely used to monitor cardiac activity
- Manual interpretation is time-consuming and requires expert knowledge
- Focus on automatic detection of abnormal cardiac activity
- Realistic conditions: noise and signal variability

STAKEHOLDERS

- Clinicians (decision support)
- Hospitals and healthcare systems
- Patients requiring long-term monitoring
- Wearable health device developers

DATASET

MIT-BIH Arrhythmia Database

Size

- 48 half-hour ECG recordings
- 47 subjects
- 110,000 labeled heartbeats

Key Features

- Two-channel ECG, 360 Hz
- Cardiologist-labeled beats
- Normal vs abnormal rhythms
- Realistic noise and variability

Machine Learning Models

Classical ML Model

- Principal Component Analysis (PCA) + Support Vector Machine (SVM)
- Hand-crafted electrocardiogram (ECG) features
- Dimensionality reduction for improved feature representation

Neural Network Model

- Regularized One-Dimensional Convolutional Neural Network (1D CNN)
- Operates directly on raw ECG signals
- Incorporates Batch Normalization and Dropout to improve generalization